Brice Peres

PhD candidate – Computer Graphics & Physical Simulation

Experience

Inria Elan team | PhD candidate – numerical study of naturally-curved fibre assemblies
Apr. 2026 – present | Grenoble, France

- Numerical modelling and simulation of bundles of naturally-curved fibres (continuation of the work below).

Inria Elan team | Pre-PhD researcher – tomographic hair-data filtering
Sept. 2025 – Mar. 2026 | Grenoble, France

- Designed filtering methods for tomographic acquisitions of human hair, in preparation of the PhD project.

Dassault Systèmes | End-of-studies internship
Sept. 2024 – Mar. 2025 | Paris, France

- Worked on the performance of one of Dassault Systèmes' main 3D engines, alongside the rendering team.

Corys | R&D Intern – Unreal Engine 5
May 2023 – Sept. 2023 | Grenoble, France

- Implemented real-time windscreen effects (rain, snow, dirt, cleaning) for a runtime train simulator on Unreal Engine 5 (HLSL).
- Worked in the R&D team led by Alain Kocelniack.

Inria | Research Intern – GPU terrain filtering
June 2022 – July 2022 | Grenoble, France

- Designed a GPU height-map filtering algorithm for terrain generalisation (advisors: Joëlle Thollot, Romain Vergne).

Selected projects

Real-time soft-body simulation | C++ / CUDA
Dec. 2024
Spring-mass soft body accelerated on the GPU with CUDA.

Progressive raytracer | C++ / OpenGL
Dec. 2024
GPU raytracer with BVH traversal, shadows, reflections, refractions, environment maps and Cook–Torrance shading with a GGX microfacet distribution. Frame accumulation for progressive refinement.

Particle physics simulation | C++
May 2023
2D/3D simulation under gravity and a Lennard-Jones potential, with hand-written vector class and spatial data structures. Ran 8 000 particles over 400 000 steps in 20 minutes on a laptop.

Black Seas – multiplayer naval game | Rust / OpenGL, team of 5
Feb. – Apr. 2023
From-scratch engine; rendering, terrain, lighting, post-processing.

Contact

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Skills

Languages
C++, Rust, C, Python, GLSL, HLSL

Graphics & GPU
OpenGL, CUDA, WebGL, Unreal Engine 5, Shadertoy

Tools
Git, LaTeX, Linux

Familiar
Java, Shell, Assembly

Languages

French
Native

English
C1 – TOEFL 100 pts

Japanese
A1

Education

ENSIMAG (M.Eng.) | École nat. sup. d'informatique et de math. appl. – Grenoble INP
2021 – 2024 | Grenoble, France

Institute of Science Tokyo | Academic exchange (formerly Tokyo Tech)
Mar. – Aug. 2024 | Tokyo, Japan

Lycée Notre-Dame de Sion | Classes préparatoires MPSI/MP
2018 – 2021 | Marseille, France

Interests

Climbing, bouldering & hiking — since 2011, with a few climbing competitions.

Computer graphics, physically based rendering, real-time simulation, Shadertoy.